

Data Analysis Project Report

E-Commerce Store Management System

Comprehensive EDA & Machine Learning Analysis using Orange Data Mining

Academic Year 2025-2026 | Database Management Course

1. Project Overview

This report comprehensively documents all data analysis steps executed on the E-Commerce Store Management System project using Orange Data Mining software. The dataset represents real e-commerce transactions from an online retail store operating across 10 Turkish cities.

1.1 Dataset Description

File: DM-project.xlsx (also used as Final_Ecommerce_data.csv)

Total Records: 22,049 rows (transactions)

Total Features: 20 columns

Notes	Type	Description	Column Name
Unique per order	Text	Unique order identifier	Order_ID
Unique per customer	Text	Customer identifier	Customer_ID
2023-01 to 2024-03	DateTime	Order date	Date
Range: 18-75	Numeric	Customer age in years	Age
Male/Female/Other	Categorical	Customer gender	Gender
10 Turkish cities	Categorical	Customer city	City
8 categories	Categorical	Product category	Product_Category
Mean: 449.70	Numeric	Price per unit	Unit_Price
Range: 1-5	Numeric	Number of items ordered	Quantity
Many zeros (no discount)	Numeric	Discount applied	Discount_Amount
Mean: 1210.69	Numeric	Final amount paid	Total_Amount
5 methods available	Categorical	Payment method used	Payment_Method
Mobile/Desktop/Tablet	Categorical	Device used to shop	Device_Type
Mean: 14.54 min	Numeric	Browsing session duration	Session_Duration_Minutes
Mean: 9 pages	Numeric	Number of pages viewed	Pages_Viewed
Target Variable	Binary	Repeat buyer indicator	Is_Returning_Customer
Range: 1-25 days	Numeric	Days to deliver order	Delivery_Time_Days
Mean: 3.90 / 5	Numeric	Customer satisfaction rating	Customer_Rating

2. Orange Workflow - Step-by-Step Analysis

The complete Orange workflow consists of 35 interconnected widgets organized into 5 major phases. The target variable is Is_Returning_Customer (binary classification: True/False).

Phase 1: Data Loading & Initial Inspection

Step 1: File Widget

What was done:

- Loaded DM-project.xlsx (22,049 records, 20 columns) into Orange
- Orange auto-detected column types: numeric, categorical, text, datetime
- Configured the file connection as the data source for the entire workflow

Why this matters:

This is the entry point for all analysis. Orange reads the file structure and creates the internal data domain that all subsequent widgets reference.

Step 2: Data Table (1)

What was done:

- Displayed all 22,049 records in tabular format
- Visually verified data loaded correctly - confirmed column names and sample values
- Observed that Pages_Viewed and Product_Category appear as duplicate columns

Insight:

Customer CUST_00001 made multiple purchases: Books (29.18), Home & Garden (506.35), Sports (1664.10) - indicating a high-value repeat customer pattern.

Step 3: Feature Statistics

What was done:

- Computed descriptive statistics for all columns simultaneously
- Identified distribution shape and missing values

Key Statistical Findings:

Interpretation	Std Dev	Max	Mean	Variable
Serves all age groups equally	11.05	75	34.97 yrs	Age

High price variance - Electronics drives this	720.09	7,900.01	449.70	Unit_Price
Low-quantity orders dominate	1.45	5	2.83	Quantity
Most orders have no discount (Q1=0)	216.62	6,538.29	59.60	Discount_Amount
Heavily right-skewed by high-value orders	2,265.76	37,852.05	1,210.69	Total_Amount
Healthy engagement time	4.86	73 min	14.54 min	Session_Duration
High variability in delivery efficiency	3.48	25	6.50 days	Delivery_Time_Days
Generally satisfied customers	1.13	5	3.90 / 5	Customer_Rating

Missing Data Assessment:

- Pages_Viewed: 4 missing values (0.018% - excellent data quality)
- Is_Returning_Customer: 4 missing values (will be imputed)
- All other columns: 0 missing values

Phase 2: Visual Exploratory Data Analysis (EDA)

Step 4: Distributions Widget

What was done:

- Generated frequency distribution histograms for all variables
- Examined distribution shape: normal / skewed / uniform
- Last examined: Delivery_Time_Days and Payment_Method

Key Findings:

- Age: Near-uniform distribution (18-75 years) - the store serves all age groups equally with no dominant segment
- Delivery_Time_Days: Uniform distribution (1-25 days) - large operational inconsistency in delivery performance
- Customer_Rating: Positively skewed toward higher ratings (mode = 4-5) - overall customer satisfaction is good
- Payment_Method: Five methods distributed nearly equally - no single dominant payment preference
- Total_Amount: Right-skewed with long tail - most transactions are small but high-value outliers exist

Step 5: Box Plot (1) - Unit_Price & Discount by Product_Category

What was done:

- Plotted box plots for Unit_Price grouped by Product_Category
- Separately plotted Discount_Amount by Payment_Method

Analysis & Interpretation:

- Electronics: Highest median price with the widest IQR - most price-variable category
- Books and Food: Narrow IQR and low prices - consistent pricing
- Outliers clearly visible in Electronics and Sports - premium products exist
- Discount_Amount median = 0 across all categories - discounts are exception not rule

Step 6: Box Plot (2) - Total_Amount by Payment_Method**What was done:**

- Compared Total_Amount distribution across all 5 payment methods

Findings:

- All 5 payment methods show similar median Total_Amount (~\$300-500)
- Credit Card and Digital Wallet show highest-value outliers
- Cash on Delivery has fewer extreme outliers - cash customers tend to order lower-value items
- No statistical difference in spending by payment method

Step 7: Box Plot (3) - Customer_Rating by Product_Category**What was done:**

- Compared satisfaction ratings across all 8 product categories

Findings:

- Customer ratings are remarkably consistent across all categories (mean ~3.9)
- No category generates significantly higher or lower satisfaction
- This means product quality/experience is uniformly managed across categories

Step 8: Box Plot (4) - Delivery_Time_Days by City**What was done:**

- Compared delivery performance across all 10 cities

Findings:

- Delivery times are uniformly distributed (1-25 days) across all cities
- No city receives significantly faster or slower delivery

- The high variability is systemic, not geographic - suggesting process issues

Step 9: Scatter Plot (1) - Unit_Price vs Total_Amount colored by Product_Category

What was done:

- Plotted Unit_Price (X) against Total_Amount (Y), colored by category

Findings & Interpretation:

- Strong positive correlation between Unit_Price and Total_Amount (expected)
- Electronics cluster in top-right: high unit price, high total
- Books and Food cluster in bottom-left: low unit price, low total
- Quantity amplifies Total_Amount - points above the trend line represent multi-unit orders
- This scatter confirms product category is the primary driver of revenue per transaction

Step 10: Correlations Widget

What was done:

- Computed Pearson correlation matrix for all numeric variables
- Top correlations displayed: Pages_Viewed(1) & Pages_Viewed(2) = 1.0

Key Correlation Findings:

- Pages_Viewed columns: $r = 1.0$ - confirmed duplicate columns in the dataset
- Product_Category columns: identical values - also duplicates to be removed
- Quantity & Total_Amount: positive correlation - more units = higher total
- Unit_Price & Total_Amount: strong positive correlation - price drives total
- Most other variable pairs: low correlation - confirming feature independence

Step 11: Scatter Plot (2) from Correlations

What was done:

- Displayed connected scatter plot showing top correlated pair: Unit_Price vs Total_Amount colored by Quantity

Insight:

The three-way relationship is clear: High Price + High Quantity = Maximum Revenue. Points in the top-right represent bulk electronics orders.

Step 12: Group By Widget

What was done:

- Grouped all 22,049 records by Customer_ID
- Computed SUM of Total_Amount per customer
- Results displayed in Data Table (1) widget

Business Value:

- This operation is equivalent to SQL Q1 (Customer Master View) - generating customer lifetime value
- Enables identification of Top 20 customers by total spending
- Key insight for loyalty program targeting

Phase 3: Customer Segmentation via Clustering

Step 13: K-Means Clustering

Configuration:

- K = 3 clusters (fixed), search range K = 2 to 8
- normalize = True (z-score normalization applied)
- max_iterations = 300, n_init = 10 (robust initialization)
- smart_init = 0 (random initialization)

What was done:

- Applied K-Means to all numeric features from the raw data
- Generated 4 Scatter Plots to interpret cluster characteristics

Scatter Plot (3): Total_Amount vs Session_Duration_Minutes

- Cluster 0 (Low-value customers): Low spending, short sessions (<10 min)
- Cluster 1 (Mid-value customers): Moderate spending, average sessions (10-20 min)
- Cluster 2 (High-value VIP customers): High spending, long sessions (>20 min)
- Key insight: Session duration positively correlates with spending - engaged browsers buy more

Scatter Plot (4): Delivery_Time_Days vs Customer_Rating

- Clusters are interleaved across delivery/rating space
- No clear cluster separation on these dimensions
- Implication: Customer segment (value tier) does not predict delivery experience
- Delivery time alone does not determine satisfaction rating

Scatter Plot (5): Unit_Price vs Quantity

- High-price products (Electronics) are purchased in lower quantities (1-2 units)

- Low-price products (Food, Books) are purchased in higher quantities (3-5 units)
- Natural price elasticity pattern confirmed in data

Scatter Plot (6): Age vs Total_Amount

- No age-based cluster separation - age does not predict spending tier
- All three clusters span all age groups uniformly (18-75 years)
- The store has no age-dominant customer segment

K-Means Conclusion:

Three customer segments identified: (1) Casual Buyers - low frequency, low value; (2) Regular Customers - moderate engagement; (3) VIP/Loyal Customers - high engagement and spend. This segmentation directly enables targeted marketing campaigns.

Phase 4: Data Preprocessing for Machine Learning

Step 14: Select Columns

What was done:

- Defined Is_Returning_Customer as the TARGET variable
- Removed meta variables: Order_ID, Customer_ID, Date (identifiers, not predictors)
- Removed duplicate columns: Product_Category(2), Pages_Viewed(2)
- Retained all relevant numeric and categorical features as inputs

Why Is_Returning_Customer as target?

Binary classification (True/False) is well-suited to the available algorithms. Predicting customer return behavior has high commercial value: it drives retention strategy, personalization, and loyalty program design.

Step 15: Impute (Missing Value Handling)

What was done:

- Addressed 4 missing values in Pages_Viewed
- Addressed 4 missing values in Is_Returning_Customer
- Applied 'Most Frequent' imputation for the target binary variable

Why this approach:

Deleting 4 records from 22,049 is negligible, but imputation is preferred for completeness. 'Most Frequent' preserves the existing class balance without introducing artificial bias.

Step 16: Data Sampler

Configuration:

- Sampling type: Fixed proportion - 70% training, 30% testing
- Stratify = True (preserves class distribution in both splits)
- Use seed = True (reproducibility)
- Output: Remaining Data (70%) flows to Preprocess and model training

Why train/test split:

Prevents overfitting assessment - the model is evaluated on data it has never seen.
Stratification ensures Is_Returning_Customer's True/False ratio is maintained in both sets.

Step 17: Preprocess Widget

Configuration:

- Normalization: Applied to all continuous numeric features
- Multinomial Treatment: Categorical encoding for SVM compatibility

Importance:

SVM is distance-based and sensitive to feature scale. Without normalization, Total_Amount (range: 6-37,852) would dominate over Age (18-75). After normalization, all features contribute equally.

Step 18: Edit Domain

What was done:

- Verified domain settings and variable type assignments
- Ensured Is_Returning_Customer is correctly set as discrete target variable
- Output flows into Outliers detection widget

Step 19: Outliers Widget

Configuration:

- Method: Isolation Forest (default)
- Contamination: 10% (expected proportion of anomalies)

What was done:

- Identified approximately 10% of records as statistical outliers
- These outlier records are labeled and passed to Test & Score as training data
- Examples of outliers: Total_Amount = 37,852 or Session_Duration = 73 minutes

Why use Outliers data for training:

The outlier records represent the edge cases the model needs to handle. Including them in training improves model robustness on unusual but real transactions.

Step 20: Rank Widget (Feature Importance)

Configuration:

- Methods: RReliefF, Univariate Regression, Information Gain Ratio, Gini Decrease
- Selected top 5 most important features

Top Ranked Features for Predicting Is_Returning_Customer:

- 1st: Session_Duration_Minutes - highest predictive power across all methods
- 2nd: Gender=Other - significant discriminating feature
- 3rd: Quantity - order quantity distinguishes return customers
- 4th: City=Kayseri - geographic feature with discriminating value
- 5th: Date-related features - temporal patterns in returning behavior

Business Interpretation:

Customers who spend more time browsing (Session_Duration) are most likely to be returning customers. This insight can directly inform UX strategy - improving site engagement converts casual visitors to loyal buyers.

Phase 5: Machine Learning Models & Evaluation

Step 21: Decision Tree Learner

Configuration:

- Max Depth = 100 (unconstrained tree depth)
- Min Leaf = 2, Min Internal = 5
- Sufficient Majority = 95%

Why Decision Tree:

Interpretable, fast, and handles both numeric and categorical features without explicit encoding. Produces human-readable IF-THEN rules. Serves as the baseline explainable model in this comparison.

Step 22: SVM Learner (Support Vector Machine)

Configuration:

- Kernel: RBF (Radial Basis Function)
- C = 1.0, Max Iterations = 100

- SVM Type: C-SVM (standard classification)

Why SVM:

SVM excels at finding optimal decision boundaries in high-dimensional spaces. RBF kernel captures non-linear relationships between features. Well-suited when feature interactions are complex.

Step 23: Random Forest Learner

Configuration:

- N Estimators = 10 trees, Max Depth = 3
- Min Samples Split = 5, Max Features = 5

Why Random Forest:

Ensemble of Decision Trees that averages predictions. More resistant to overfitting than a single tree. Provides feature importance rankings consistent with the Rank widget results.

Step 24: Test and Score

Configuration:

- Cross Validation: 3 folds, 5 repetitions
- All 3 learners evaluated simultaneously on the same data
- Training data: Outlier-labeled records from Outliers widget

Evaluation Metrics Explained:

Ideal Value	What It Measures	Metric
1.0	Overall correct predictions / total predictions	CA (Accuracy)
1.0	Area under the ROC curve (discrimination ability)	AUC
1.0	Harmonic mean of Precision and Recall	F1 Score
1.0	Of predicted positives, how many are correct	Precision
1.0	Of actual positives, how many were found	Recall

Step 25: Confusion Matrix

What was done:

- Displayed confusion matrix for Random Forest model
- Selected learner: Random Forest (index 2 in learner list)

Confusion Matrix Terminology:

- True Positive (TP): Returning customer correctly predicted as returning
- True Negative (TN): New customer correctly predicted as new
- False Positive (FP): New customer incorrectly predicted as returning
- False Negative (FN): Returning customer incorrectly predicted as new
- In the context of customer retention, FN is the costlier error (missed opportunity for retention marketing)

Step 26: Scatter Plot (7) from Rank Output

What was done:

- Visualized top-ranked features after Rank widget processing
- Date vs Session_Duration_Minutes colored by Is_Returning_Customer

Interpretation:

This scatter confirms the Rank findings visually - returning customers (True) tend to cluster at higher session durations. The temporal dimension (Date) shows consistent patterns throughout the analysis period.

Step 27: Predictions Widget

What was done:

- Applied trained Random Forest model to the holdout test set (30% of data)
- Displayed actual vs predicted Is_Returning_Customer side by side
- Showed prediction probability for each customer record

Step 28: Save Data

Configuration:

- Output file: Final_Predictions_DM.Excel.tab
- Format: Tab-separated values
- Content: All original features + model predictions + probabilities

This final step exports actionable results that can be imported into SQL Server or Excel for business use.

3. Key Insights & Findings Summary

3.1 Business Intelligence Discoveries

Business Action	Finding	Analysis Area
Tailor marketing to 3 distinct customer tiers	3 Customer segments identified via K-Means	Customer Segmentation
Focus inventory investment on Electronics	Electronics drives highest revenue per transaction	Product Revenue
Build UX features to increase session length	Session duration is top predictor of loyalty	Retention Driver
Investigate delivery process inconsistency	Delivery varies wildly: 1 to 25 days	Operational Issue
Target all age groups equally in campaigns	No age group dominates spending behavior	Demographics
Audit data pipeline for duplicate columns	Pages_Viewed and Product_Category duplicated	Data Quality Issue

3.2 Workflow Quality Assessment

Strengths of the Implemented Workflow:

- Logical sequential flow: Exploration -> Clustering -> Preprocessing -> Modeling
- EDA performed thoroughly before jumping to machine learning
- Missing value imputation performed before model training
- Proper train/test split with stratification
- Feature normalization applied before SVM (correct practice)
- Three algorithms compared simultaneously for robust model selection
- Outlier detection integrated into the pipeline
- Rank widget used to understand feature importance before modeling
- Predictions saved to file for downstream business use

Recommendations for Enhancement:

- Remove duplicate columns (Pages_Viewed.1, Product_Category.1) at ingestion
- Add ROC Curve widget for visual model comparison
- Consider expanding K-Means to test K=4 or K=5 for finer segmentation
- Add a Calibration Plot to verify probability outputs from models

3.3 Data Quality Summary

- Overall data quality: Excellent - 22,049 records with near-zero missing values
- Missing rate: $4/22,049 = 0.018\%$ - well below the 5% acceptable threshold

- Date coverage: January 2023 to March 2024 - 15 months of data
- Geographic coverage: 10 major Turkish cities
- Data issue detected: Duplicate columns for Pages_Viewed and Product_Category

4. Conclusion

This project successfully implements a complete end-to-end data analysis pipeline for an E-Commerce Management System using Orange Data Mining. The workflow processes 22,049 real transactional records through 35 interconnected widgets across 5 analytical phases.

The analysis delivers four major outcomes:

- Comprehensive EDA revealing customer behavior patterns, product performance, and operational metrics
- Customer segmentation into 3 actionable business tiers via K-Means clustering
- Machine learning model (best: Random Forest) predicting customer return probability
- Exportable predictions ready for integration with SQL Server business intelligence

Primary objective achieved: Complete EDA + ML Classification + Actionable Business Insights.

Report prepared from Orange Workflow analysis (Final_DM_ORANGE.ows) and dataset (DM-project.xlsx) | Academic Year 2025-2026